

Parent Guide — middle / module-05-dot-addition

IKS Curriculum

Table of Contents

Parent Guide — Module 5: Bindu Paddhati (Dot Method)

Your child's age: Middle (Grades 6–8) · **Module length:** 10 sessions × 45 min

What your child is learning

The dot method — a mental-arithmetic technique for adding long columns of numbers fast. Instead of adding digit-by-digit, your child learns to *scan* a column for pairs summing to 10 and mark each “completion” with a dot. The dot count becomes the carry.

Baseline → retest: On Day 1, your child times themselves adding a standard 5-addend × 5-digit sum using their normal method. On Day 5, they retest the same problem using the dot method. The visible speedup (50%+ for most students) is the motivation.

Mental shortcuts (Day 7): Adding 9 to anything = add 10, subtract 1. ($33 + 9 = 33 + 10 - 1 = 42$.) This is Tirthaji's 7th sūtra, *Sankalana Vyavakalanābhyām* — “by addition and by subtraction.”

A note on “Vedic mathematics”

Important honest framing: - The dot method is a real, useful Indian pedagogical technique. - It is **not directly from any Vedic text**. It is a 20th-century teaching tool that draws on the spirit of Bharati Krishna Tirthaji's 16 sūtras (published 1965, posthumous). - The math is real. Tirthaji's sūtras are real and culturally significant. The dot method itself is more recent than its name suggests.

Your child will explore this honest framing on Day 6 (“Why it works”). At Senior band they'll read the scholarly debate (Dani 1993) about Tirthaji's claims.

What to expect this fortnight

Daily warm-up: 1 minute of “make-10 fluency” — instant recall of 10-pairs.

Speed drills: paired practice, self-timed. No public leaderboards — your child races themselves, not classmates.

Project (Days 8–10): Your child chooses one of: - A 60-second teaching video (Grade 4 audience). - A printed 20-problem workbook with worked solutions. - A comparative speed-test data project (collect times from 5–10 family/friends).

How to help

Helpful: - For the speed-test data project: be a willing test subject. Let your child time your standard-algorithm addition vs. their dot-method addition. - 5-minute home drill: call out single digits, your child answers the 10-partner instantly. Then 2-digit add-9 problems. - Try the dot method yourself on a 4×4-digit sum. Notice the scanning skill.

Not helpful: - Insisting their teacher’s “old way” is wrong. Both methods give the same answer; the dot method is faster on multi-addend columns. - Doing their project. Even the workbook has to be in their handwriting / their drawings.

What success looks like

By end of module, your child can: - Add a 4-addend × 4-digit column in under 60 seconds using the dot method. - Do mental “+9, +19, +29” instantly. - Explain *why* the dot method works (a dot = 10 = 1 carried). - Distinguish the dot method (technique) from Tirthaji’s sūtras (named formulas).

Citation standard

Projects must cite the source PDF (e.g. *Vedic Mathematics Batch1.pdf*, §5.4). The full citation chain is documented in the module’s `sources.md`.

Want to go deeper?

Senior band (Grade 9–12) generalises the dot method to base- n arithmetic AND tackles the historicity debate in depth. Preview: `curriculum/senior/module-05-dot-addition/README.md`.

10-Day Lesson Plan

Module 5 (Middle) — 10-Day Lesson Plan

At a glance

Day	Session title	Lesson file	Activity / assessment
1	The Hook — How fast can you add?	<code>days/day-01-hook.md</code>	Baseline timed test
2	Making Tens	<code>days/day-02-making-tens.md</code>	Pair drills with the “10-pair” cards
3	Single Column with Dots	<code>days/day-03-single-</code>	Formative quiz

Day	Session title	Lesson file	Activity / assessment
		column.md	
4	Multiple Columns	days/day-04-multiple-columns.md	activities/activity-01-column-drill.md
5	Mid-module Check	days/day-05-mid-module-check.md	quizzes/formative.md Part B
6	Why It Works	days/day-06-why-it-works.md	Reflection paragraph
7	Mental Math Patterns	days/day-07-mental-patterns.md	activities/activity-02-near-base.md
8	Project Session 1	days/day-08-project-session-1.md	Project draft
9	Project Session 2	days/day-09-project-session-2.md	Project draft
10	Final Assessment	days/day-10-final-assessment.md	quizzes/summative.md + project

Pacing notes

- Day 1 establishes a baseline speed (each student times their own standard-algorithm addition of a 4×4-digit sum). Day 5 retests the SAME problem set with the dot method; the visible speedup is the module’s motivation.
- Days 2–4 are the core. Compressed schedule: combine Days 2+3 into one period if needed.
- Day 6 (“Why it works”) can be pushed to Day 7+ if the class is enjoying the drills.
- Project (Days 8–10) is shorter than Panchabhuta’s because the artefact options are more constrained.

Daily warm-up (recurring)

Every class starts with **1 minute of “make-10 fluency”**: teacher calls out single digits; students call back the partner that makes 10. (8 → 2, 6 → 4, 9 → 1, etc.) This is the foundational skill.

What students leave the module with

- Fluency at scanning a column of numbers and finding 10-pairs.
- Confidence adding 4-digit, 5-addend columns in their head or with paper.
- Awareness that “Indian Math” is not a single ancient block — it includes 20th-century pedagogical innovations like this one.
- One practice artefact (their project) they can share with younger students.

Homework

Homework — Module 5 Dot Addition (Middle)

Day 1 → Day 2

Task: Practice the 10-pair reflex. Do all 10 in under 15 seconds: $8+?=10$ $3+?=10$ $6+?=10$ $4+?=10$ $9+?=10$ $1+?=10$ $7+?=10$ $5+?=10$ $2+?=10$ $0+?=10$

Time: 5 min.

Day 2 → Day 3

Task: Have a parent or sibling call out 20 single digits in 30 seconds; you say the 10-partner for each. Track how many you got right.

Time: 5 min.

Day 3 → Day 4

Task: Solve the 10 column problems on the back of today's worksheet using the dot method. Self-check against standard addition.

Time: 15 min.

Day 4 → Day 5

Task: Find any column-addition problem in any textbook. Solve it BOTH ways (standard + dot method). Time yourself for each. Bring tomorrow.

Time: 10 min.

Day 5 → Day 6

Task: Submit your project plan paragraph in writing (4–6 sentences).

Time: 15 min.

Day 6 → Day 7

Task: Find ONE example in your daily life where you naturally “complete to 10” without thinking (rounding a bill, counting change, adding minutes to a clock). Write 2 sentences.

Time: 10 min.

Day 7 → Day 8

Task: 20 mental near-base additions. Do them in under 60 seconds. List on the back of today's worksheet. Bring ALL project materials tomorrow.

Time: 5 min (plus material gathering).

Day 8 → Day 9

Task: Finish at home what you didn't finish in class. Tomorrow is for refining, not starting.

Time: capped at 30 min.

Day 9 → Day 10

Task: Final touches. Make sure you have: the artefact, your script/notes, your 3-min presentation timed in your head.

Time: 15 min.